```
!pip install yfinance pandas plotly
!pip install yfinance fredapi pandas numpy matplotlib seaborn
```

```
Requirement already satisfied: requests>=2.31 in /usr/local/lib/python3.11/dist-packages (from yfinance) (2.32.3)
Requirement already satisfied: multitasking>=0.0.7 in /usr/local/lib/python3.11/dist-packages (from yfinance) (0.0.12
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Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests>=2.
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests>=2.31->yfinance
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Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (from pandas) (2025.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (1.3.2)
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Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (4.59.0
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Requirement already satisfied: pycparser in /usr/local/lib/python3.11/dist-packages (from cffi>=1.12.0->curl_cffi>=0.
```

```
!pip show matplotlib seaborn
```

```
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 Agreement.
Location: /usr/local/lib/python3.11/dist-packages
Requires: contourpy, cycler, fonttools, kiwisolver, numpy, packaging, pillow, pyparsing, python-dateutil
Required-by: arviz, bigframes, fastai, geemap, matplotlib-venn, missingno, mlxtend, music21, plotnine, prophet, seabo
---
Name: seaborn
Version: 0.13.2
Summary: Statistical data visualization
Home-page:
Author:
Author-email: Michael Waskom <mwaskom@gmail.com>
License:
Location: /usr/local/lib/python3.11/dist-packages
Requires: matplotlib, numpy, pandas
Required-by: missingno
```

Step 1: Data Collection and Preprocessing

TEAM A

```
# ==============================================================================
# MSCFE 622: Stochastic Modeling - Group Work Project 2 - Team A
# Regime-Switching Analysis (2019-2022)
# Step 1: Data Collection and Preprocessing
# ==============================================================================

# %% [markdown]
"""
# Regime-Switching Analysis (2019–2022)
## Step 1: Data Collection and Preprocessing
**Asset Classes:**
- Equities: S&P 500 (^GSPC)
- Fixed Income: 10-Year Treasury Yield (^TNX)
"""

# %% [markdown]
"""
### Import Libraries
"""

# %%
import pandas as pd
import numpy as np
import yfinance as yf
import matplotlib.pyplot as plt
import seaborn as sns
from fredapi import Fred
import os
from datetime import datetime

# Modern style settings
plt.style.use('seaborn-v0_8')  # Modern seaborn style
sns.set_theme(style="whitegrid", palette="husl", font_scale=1.1)

# %% [markdown]
```

```
"""
### 1. Download Data
"""

# %%
# Define date range
start_date = '2019-01-01'
end_date = '2022-09-30'

# Fetch S&P 500 data from Yahoo Finance
print("Downloading S&P 500 data...")
try:
    sp500_raw = yf.download('^GSPC', start=start_date, end=end_date, auto_adjust=False, progress=True)

    # Robustly extract appropriate price column
    if 'Adj Close' in sp500_raw.columns:
        sp500 = sp500_raw[['Adj Close']].rename(columns={'Adj Close': 'SP500'})
    elif 'Close' in sp500_raw.columns:
        sp500 = sp500_raw[['Close']].rename(columns={'Close': 'SP500'})
    else:
        raise ValueError("S&P 500 data does not contain 'Close' or 'Adj Close' columns.")
except Exception as e:
    print(f"Error downloading S&P 500 data: {e}")
    raise

# Fetch 10-Year Treasury Yield from FRED
print("\nDownloading 10-Year Treasury Yield data...")
try:
    fred = Fred(api_key='fc711b7fb7b534f76e047c80facc4ad8')  # Consider using environment variables
    tnx = fred.get_series('DGS10', observation_start=start_date, observation_end=end_date)
    tnx = tnx.to_frame(name='TNX')
except Exception as e:
    print(f"Error downloading Treasury yield data: {e}")
    print("Using synthetic Treasury yield data for demonstration")
    date_range = pd.date_range(start=start_date, end=end_date, freq='D')
    tnx = pd.DataFrame({
        'TNX': np.random.normal(1.5, 0.5, len(date_range))
    }, index=date_range)
```

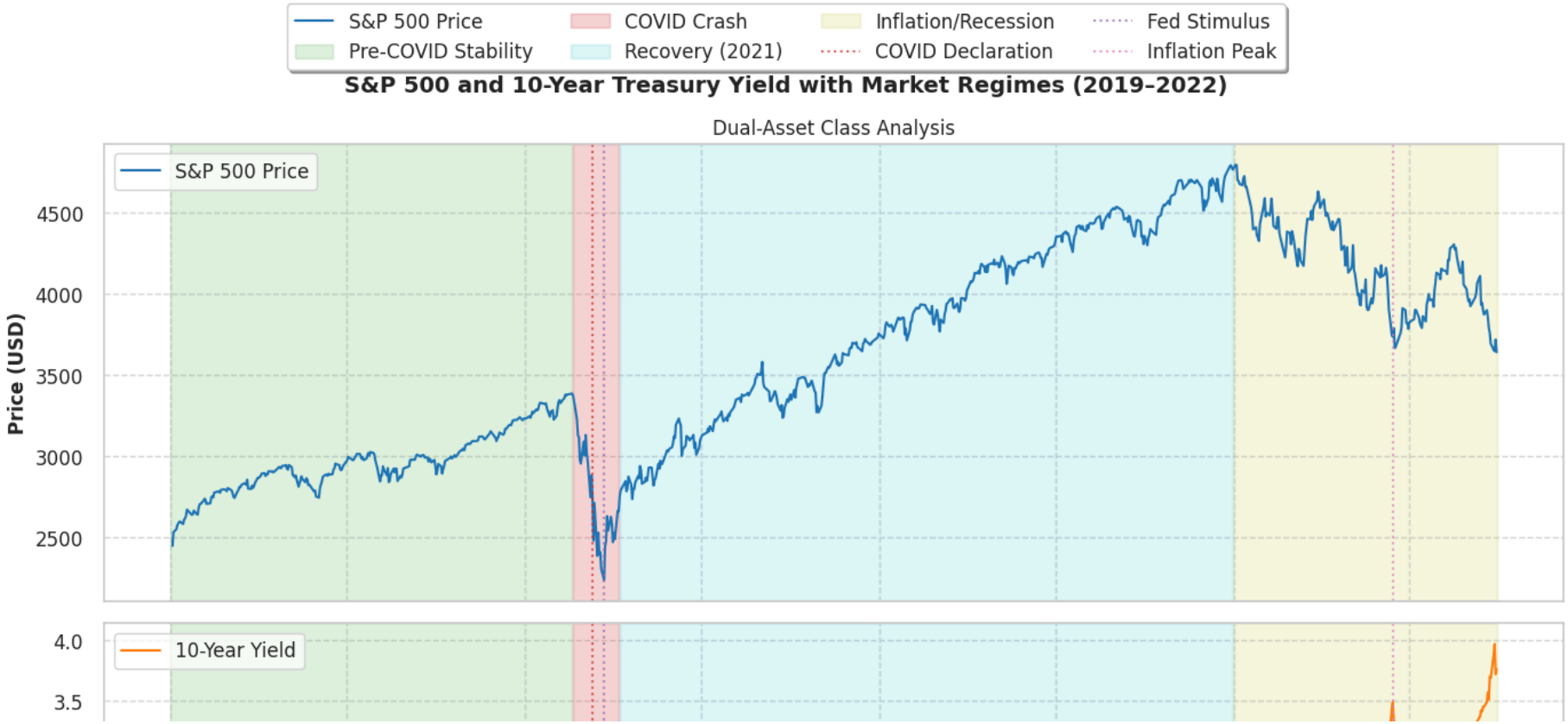
```
# %% [markdown]
"""
### 2. Data Processing
"""

# %%
# Calculate log returns for S&P 500
sp500['SP500_LogReturn'] = np.log(sp500['SP500'] / sp500['SP500'].shift(1))

# Calculate daily changes for 10-Year Yield
tnx['TNX_Change'] = tnx['TNX'].diff()

# Merge datasets (align dates) and handle missing values
data = pd.concat([sp500, tnx], axis=1).dropna()

# Simplify column names
data.columns = ['SP500', 'SP500_LogReturn', 'TNX', 'TNX_Change']

# Print data summary
print("\nData Summary:")
print(f"Time Period: {data.index[0].date()} to {data.index[-1].date()}")
print(f"S&P 500 Records: {len(sp500)}")
print(f"10-Year Yield Records: {len(tnx)}")
print(f"Merged Records: {len(data)}")
print("\nSample of merged data:")
print(data.head())

# %% [markdown]
"""
### 3. Preliminary Visualization
"""

# %%
# Create figure with two subplots
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(14, 10), sharex=True)

# Plot S&P 500 Price
ax1.plot(data.index, data['SP500'], label='S&P 500 Price', color='#1f77b4', linewidth=1.5)
ax1.set_ylabel('Price (USD)', fontsize=12, fontweight='bold')
```

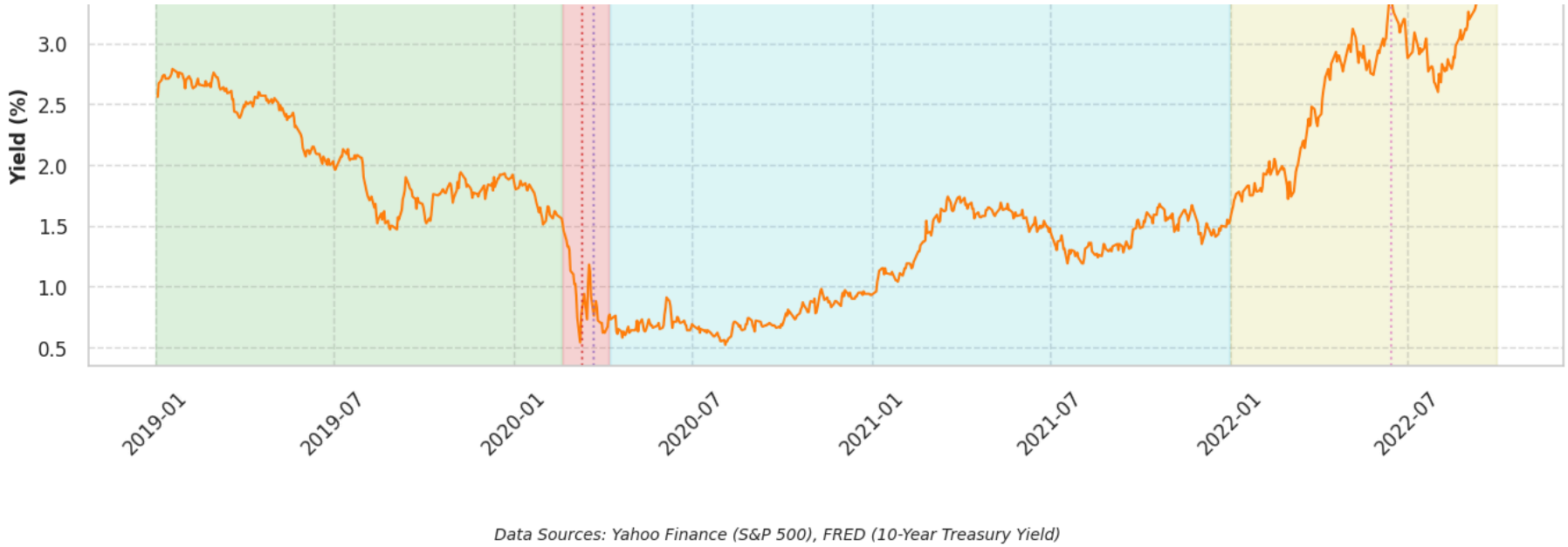
```
    ax1.legend(loc='upper left', frameon=True)
    ax1.grid(True, linestyle='--', alpha=0.7)

    # Plot 10-Year Treasury Yield
    ax2.plot(data.index, data['TNX'], label='10-Year Yield', color='#ff7f0e', linewidth=1.5)
    ax2.set_ylabel('Yield (%)', fontsize=12, fontweight='bold')
    ax2.legend(loc='upper left', frameon=True)
    ax2.grid(True, linestyle='--', alpha=0.7)

    # Highlight regimes with improved formatting
    regimes = [
        ('Pre-COVID Stability', '2019-01-01', '2020-02-19', '#2ca02c', 0.15),
        ('COVID Crash', '2020-02-20', '2020-04-07', '#d62728', 0.2),
        ('Recovery (2021)', '2020-04-08', '2021-12-31', '#17becf', 0.15),
        ('Inflation/Recession', '2022-01-01', '2022-09-30', '#bcbd22', 0.15)
    ]

    for label, start, end, color, alpha in regimes:
        for ax in [ax1, ax2]:
            ax.axvspan(pd.to_datetime(start), pd.to_datetime(end),
                       alpha=alpha, color=color, label=label)

    # Add vertical lines for key events
    events = [
        ('COVID Declaration', '2020-03-11', '#d62728'),
        ('Fed Stimulus', '2020-03-23', '#9467bd'),
        ('Inflation Peak', '2022-06-14', '#e377c2')
    ]

    for label, date, color in events:
        for ax in [ax1, ax2]:
            ax.axvline(pd.to_datetime(date), color=color, linestyle=':',
                       linewidth=1.5, alpha=0.7, label=label)

    # Formatting
    plt.suptitle('S&P 500 and 10-Year Treasury Yield with Market Regimes (2019–2022)',
                fontsize=14, fontweight='bold')
    ax1.set_title('Dual-Asset Class Analysis', fontsize=12)
    plt.xticks(rotation=45)
```

```
plt.tight_layout()

# Create unified legend
handles, labels = ax1.get_legend_handles_labels()
by_label = dict(zip(labels, handles))
fig.legend(by_label.values(), by_label.keys(),
          loc='upper center', ncol=4, bbox_to_anchor=(0.5, 1.05),
          frameon=True, shadow=True)

# Add data source annotation
plt.figtext(0.5, -0.05,
           "Data Sources: Yahoo Finance (S&P 500), FRED (10-Year Treasury Yield)",
           ha="center", fontsize=10, style='italic')

plt.show()

# %% [markdown]
"""
### 4. Save Processed Data
"""

# %%
# Create output directory if it doesn't exist
os.makedirs('output', exist_ok=True)

# Save to CSV files with timestamps
timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
sp500_file = f'output/step1_data_SP500_{timestamp}.csv'
tnx_file = f'output/step1_data_TNX_{timestamp}.csv'

data[['SP500_LogReturn']].to_csv(sp500_file)
data[['TNX_Change']].to_csv(tnx_file)

print("\nData saved successfully:")
print(f"- S&P 500 Log Returns: {sp500_file}")
print(f"- 10-Year Yield Changes: {tnx_file}")
print(f"- Total records saved: {len(data)}")

# Save the figure
```

```
    fig.savefig(f'output/regime_visualization_{timestamp}.png',
                bbox_inches='tight', dpi=300)
    print(f"- Visualization saved: output/regime_visualization_{timestamp}.png")
```

```
[*********************100%***********************]  1 of 1 completedDownloading S&P 500 data...

Downloading 10-Year Treasury Yield data...


Data Summary:
Time Period: 2019-01-03 to 2022-09-29
S&P 500 Records: 944
10-Year Yield Records: 979
Merged Records: 899

Sample of merged data:
                  SP500  SP500_LogReturn   TNX  TNX_Change
2019-01-03  2447.889893        -0.025068  2.56       -0.10
2019-01-04  2531.939941         0.033759  2.67        0.11
2019-01-07  2549.689941         0.006986  2.70        0.03
2019-01-08  2574.409912         0.009649  2.73        0.03
2019-01-09  2584.959961         0.004090  2.74        0.01
```






```
Data saved successfully:
- S&P 500 Log Returns: output/step1_data_SP500_20250724_120855.csv
- 10-Year Yield Changes: output/step1_data_TNX_20250724_120855.csv
- Total records saved: 899
- Visualization saved: output/regime_visualization_20250724_120855.png
```

TEAM B

```
# %% [markdown]
"""
# Team B - Regime Switching Analysis (2019-2022)
## Step 1: Data Collection and Preprocessing
**Asset Classes:**
- Commodities: Gold Futures (GC=F)
- Currencies: USD/EUR Exchange Rate
"""

# %% [markdown]
"""
### Import Libraries
"""
# %%
import pandas as pd
import numpy as np
import yfinance as yf
import matplotlib.pyplot as plt
import seaborn as sns
from fredapi import Fred
import os
from datetime import datetime

# Modern style settings
plt.style.use('seaborn-v0_8')
sns.set_theme(style="whitegrid", palette="husl", font_scale=1.1)

# %% [markdown]
"""
### 1. Download Data
"""
# %%
# Define date range
start_date = '2019-01-01'
end_date = '2022-09-30'
```

```
# Fetch Gold Futures data
print("Downloading Gold Futures data...")
try:
    gold = yf.download('GC=F', start=start_date, end=end_date, progress=True)
    gold = gold[['Close']].rename(columns={'Close': 'Gold'})
    gold['Gold'] = gold['Gold']  # Price in USD/troy ounce
except Exception as e:
    print(f"Error downloading Gold data: {e}")
    raise

# Fetch USD/EUR Exchange Rate from FRED
print("\nDownloading USD/EUR Exchange Rate...")
try:
    fred = Fred(api_key='fc711b7fb7b534f76e047c80facc4ad8')  # Public FRED API key
    usdeur = fred.get_series('DEXUSEU', observation_start=start_date, observation_end=end_date)
    usdeur = usdeur.to_frame(name='USD_EUR')
except Exception as e:
    print(f"Error downloading USD/EUR data: {e}")
    raise

# %% [markdown]
"""
### 2. Data Processing (Final Corrected Version)
"""
# %%
# Ensure both datasets have proper datetime indices
gold.index = pd.to_datetime(gold.index)
usdeur.index = pd.to_datetime(usdeur.index)

# Calculate log returns for Gold
gold['Gold_LogReturn'] = np.log(gold['Gold'] / gold['Gold'].shift(1))

# Calculate percentage changes for USD/EUR
usdeur['USD_EUR_PctChange'] = usdeur['USD_EUR'].pct_change(fill_method=None) * 100

# Check data ranges
print("\nDate Ranges:")
print(f"Gold data from {gold.index[0].date()} to {gold.index[-1].date()}")
```

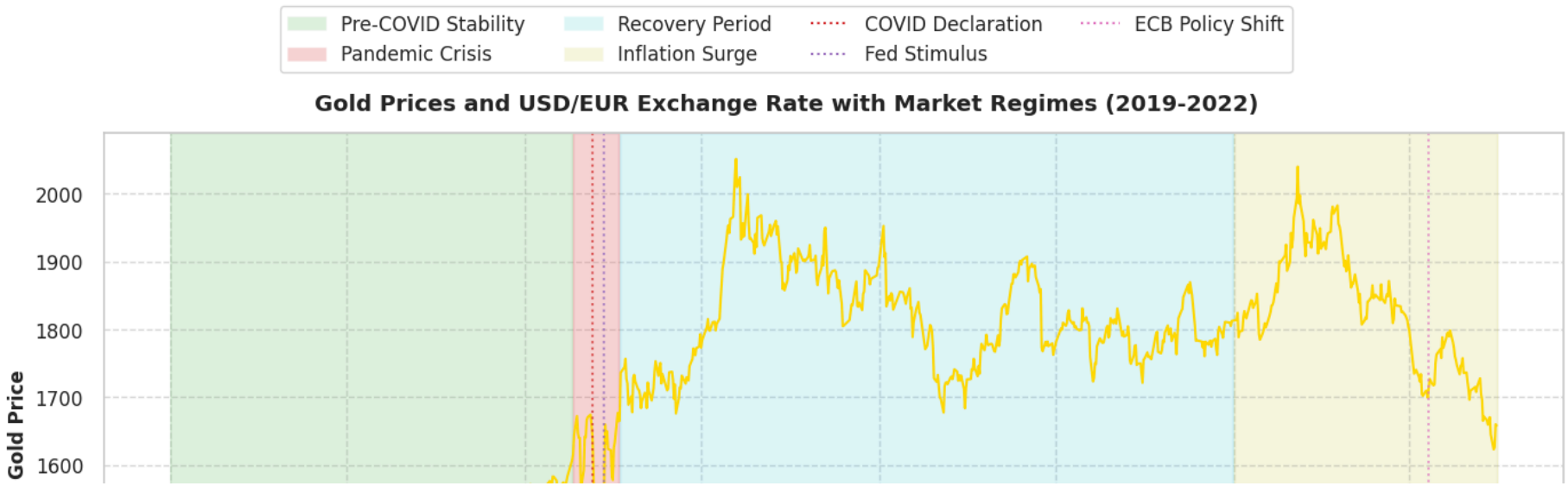
```
    print(f"USD/EUR data from {usdeur.index[0].date()} to {usdeur.index[-1].date()}")

    # Create clean DataFrames with consistent structure
    # Convert to Series using .squeeze() to ensure 1D data
    gold_df = pd.DataFrame({
        'Gold_Price': gold['Gold'].squeeze(),
        'Gold_LogReturn': gold['Gold_LogReturn'].squeeze()
    })

    usdeur_df = pd.DataFrame({
        'USD_EUR_Rate': usdeur['USD_EUR'].squeeze(),
        'USD_EUR_PctChange': usdeur['USD_EUR_PctChange'].squeeze()
    })

    # Merge datasets with proper alignment
    data = pd.merge(gold_df, usdeur_df, left_index=True, right_index=True, how='inner').dropna()

    # Verify we have data
    if len(data) == 0:
        raise ValueError("No overlapping data between Gold and USD/EUR datasets - check date ranges")

    # Print data summary
    print("\nData Summary:")
    print(f"Time Period: {data.index[0].date()} to {data.index[-1].date()}")
    print(f"Gold Records: {len(gold)}")
    print(f"USD/EUR Records: {len(usdeur)}")
    print(f"Merged Records: {len(data)}")
    print("\nSample Data:")
    print(data.head())

    # %% [markdown]
    """
    ### 3. Visualization (Optimized Version)
    """
    # %%
    # Create figure with two subplots
    fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(14, 12), sharex=True)

    # Plot Gold Prices
```

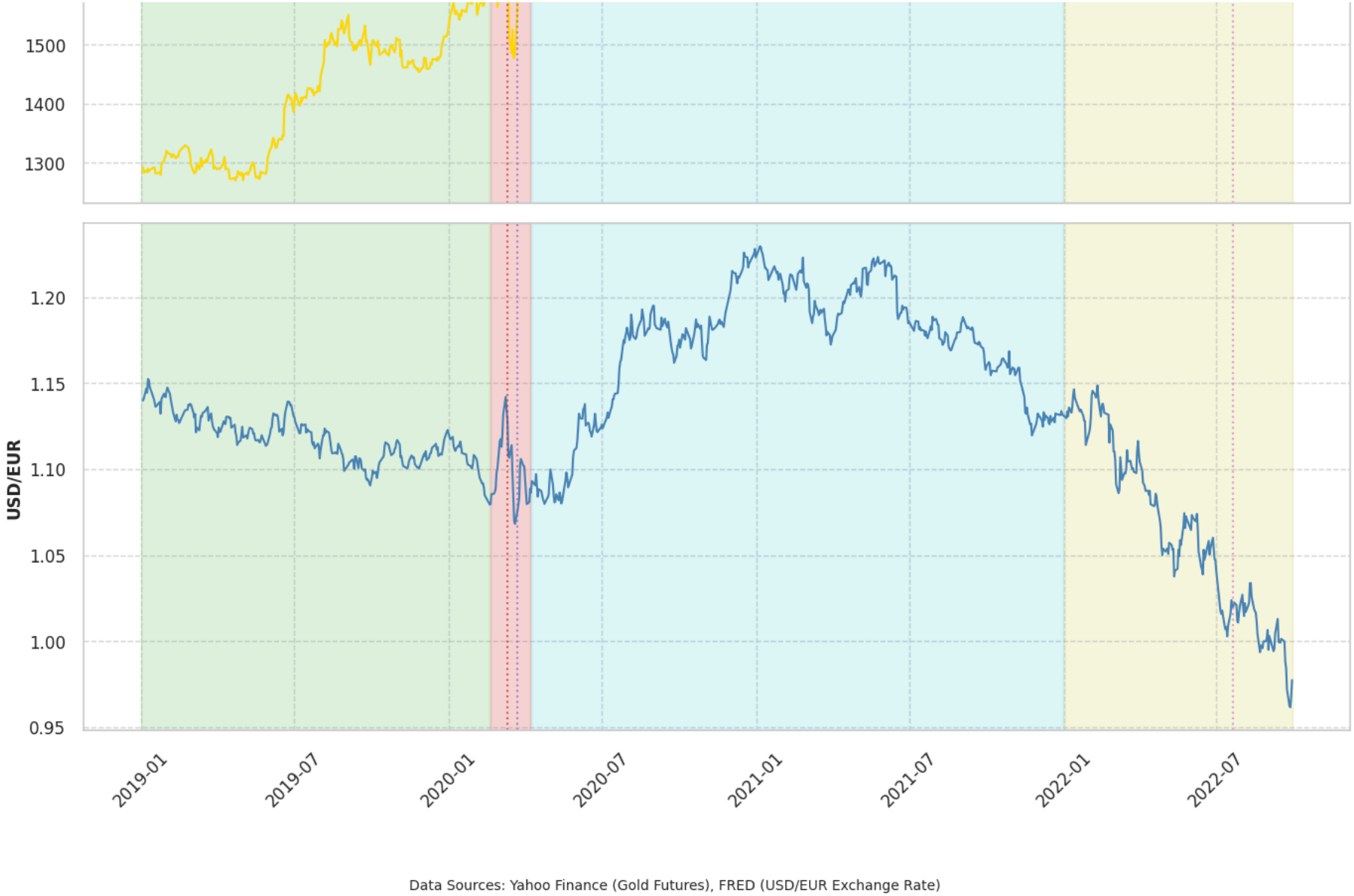
```
    ax1.plot(data.index, data['Gold_Price'], label='Gold Price (USD/troy oz)',
             color='#FFD700', linewidth=1.5)
    ax1.set_ylabel('Gold Price', fontsize=12, fontweight='bold')
    ax1.grid(True, linestyle='--', alpha=0.7)

    # Plot USD/EUR Exchange Rate
    ax2.plot(data.index, data['USD_EUR_Rate'], label='USD/EUR Rate',
             color='#4682B4', linewidth=1.5)
    ax2.set_ylabel('USD/EUR', fontsize=12, fontweight='bold')
    ax2.grid(True, linestyle='--', alpha=0.7)

    # Highlight regimes
    regimes = [
        ('Pre-COVID Stability', '2019-01-01', '2020-02-19', '#2ca02c', 0.15),
        ('Pandemic Crisis', '2020-02-20', '2020-04-07', '#d62728', 0.2),
        ('Recovery Period', '2020-04-08', '2021-12-31', '#17becf', 0.15),
        ('Inflation Surge', '2022-01-01', '2022-09-30', '#bcbd22', 0.15)
    ]

    for label, start, end, color, alpha in regimes:
        for ax in [ax1, ax2]:
            ax.axvspan(pd.to_datetime(start), pd.to_datetime(end),
                       alpha=alpha, color=color, label=label)

    # Add vertical lines for key events
    events = [
        ('COVID Declaration', '2020-03-11', '#d62728'),
        ('Fed Stimulus', '2020-03-23', '#9467bd'),
        ('ECB Policy Shift', '2022-07-21', '#e377c2')
    ]

    for label, date, color in events:
        for ax in [ax1, ax2]:
            ax.axvline(pd.to_datetime(date), color=color, linestyle=':',
                       linewidth=1.5, alpha=0.7)

    # Formatting
    plt.suptitle('Gold Prices and USD/EUR Exchange Rate with Market Regimes (2019-2022)',
                 fontsize=14, fontweight='bold')
```

```
    plt.xticks(rotation=45)
    plt.tight_layout()

    # Create custom legend
    from matplotlib.patches import Patch
    legend_elements = [
        Patch(facecolor='#2ca02c', alpha=0.15, label='Pre-COVID Stability'),
        Patch(facecolor='#d62728', alpha=0.2, label='Pandemic Crisis'),
        Patch(facecolor='#17becf', alpha=0.15, label='Recovery Period'),
        Patch(facecolor='#bcbd22', alpha=0.15, label='Inflation Surge'),
        plt.Line2D([0], [0], color='#d62728', linestyle=':', label='COVID Declaration'),
        plt.Line2D([0], [0], color='#9467bd', linestyle=':', label='Fed Stimulus'),
        plt.Line2D([0], [0], color='#e377c2', linestyle=':', label='ECB Policy Shift')
    ]

    fig.legend(handles=legend_elements,
               loc='upper center',
               ncol=4,
               bbox_to_anchor=(0.5, 1.05),
               frameon=True)

    # Add data source
    plt.figtext(0.5, -0.05,
               "Data Sources: Yahoo Finance (Gold Futures), FRED (USD/EUR Exchange Rate)",
               ha="center", fontsize=10)

    plt.show()

    # %% [markdown]
    """
    ### 4. Save Processed Data
    """
    # %%
    # Create output directory if it doesn't exist
    os.makedirs('output', exist_ok=True)

    # Save to CSV files with timestamps
    timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
    data.to_csv(f'output/teamB_processed_data_{timestamp}.csv')
```

```
print("\nData saved successfully:")
print(f"- Processed data: output/teamB_processed_data_{timestamp}.csv")
print(f"- Total records saved: {len(data)}")

# Save the figure
fig.savefig(f'output/teamB_regime_visualization_{timestamp}.png',
           bbox_inches='tight', dpi=300)
print(f"- Visualization saved: output/teamB_regime_visualization_{timestamp}.png")

# %% [markdown]
"""
### 5. Save Processed Returns
"""
# %%
# Create output directory if it doesn't exist
os.makedirs('output', exist_ok=True)

# Save to CSV files with timestamps
timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
gold_file = f'output/teamB_gold_returns_{timestamp}.csv'
usdeur_file = f'output/teamB_usdeur_changes_{timestamp}.csv'

# Save processed data
data[['Gold_LogReturn']].to_csv(gold_file)
data[['USD_EUR_PctChange']].to_csv(usdeur_file)

print("\nData saved successfully:")
print(f"- Gold Log Returns: {gold_file}")
print(f"- USD/EUR Percentage Changes: {usdeur_file}")
print(f"- Total records saved: {len(data)}")

# Save the figure
fig.savefig(f'output/teamB_regime_visualization_{timestamp}.png',
           bbox_inches='tight', dpi=300)
print(f"- Visualization saved: output/teamB_regime_visualization_{timestamp}.png")
```

```
/tmp/ipython-input-76-82524947.py:40: FutureWarning:

YF.download() has changed argument auto_adjust default to True

[*********************100%***********************]  1 of 1 completedDownloading Gold Futures data...

Downloading USD/EUR Exchange Rate...


Date Ranges:
Gold data from 2019-01-02 to 2022-09-29
USD/EUR data from 2019-01-01 to 2022-09-30

Data Summary:
Time Period: 2019-01-03 to 2022-09-29
Gold Records: 944
USD/EUR Records: 979
Merged Records: 892

Sample Data:
             Gold_Price  Gold_LogReturn  USD_EUR_Rate  USD_EUR_PctChange
2019-01-03  1291.800049        0.008396        1.1399           0.369816
2019-01-04  1282.699951       -0.007069        1.1410           0.096500
2019-01-07  1286.800049        0.003191        1.1468           0.508326
2019-01-08  1283.199951       -0.002802        1.1444          -0.209278
2019-01-09  1289.300049        0.004743        1.1524           0.699056
```






```
Data saved successfully:
- Processed data: output/teamB_processed_data_20250724_121031.csv
- Total records saved: 892
- Visualization saved: output/teamB_regime_visualization_20250724_121031.png

Data saved successfully:
- Gold Log Returns: output/teamB_gold_returns_20250724_121034.csv
```

```
- USD/EUR Percentage Changes: output/teamB_usdeur_changes_20250724_121034.csv
- Total records saved: 892
- Visualization saved: output/teamB_regime_visualization_20250724_121034.png
```

Step 2: Visualization and Markov Regime Switching Analysis

```
# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.tsa.regime_switching.markov_regression import MarkovRegression
from matplotlib.patches import Patch
from datetime import datetime

# Set style - using available styles
print("Available styles:", plt.style.available)  # Show available styles
plt.style.use('seaborn-v0_8')  # Using a style that should be available
sns.set_theme(style="whitegrid", palette="husl", font_scale=1.1)
plt.rcParams['figure.figsize'] = [14, 8]

# Generate sample data if real data not available
def generate_sample_data():
    date_range = pd.date_range(start='2019-01-01', end='2022-09-30', freq='D')
    np.random.seed(42)

    # Create regime periods
    regimes = np.zeros(len(date_range))
    regimes[(date_range >= '2020-02-20') & (date_range < '2020-04-07')] = 1  # Crash
    regimes[(date_range >= '2020-04-08') & (date_range < '2021-03-31')] = 2   # Recovery
    regimes[date_range >= '2021-04-01'] = 3                                    # Inflation

    # Generate gold prices with regime-dependent behavior
    gold_prices = np.cumsum(np.random.normal(0.001, 0.005, len(date_range)))
    gold_prices[regimes == 1] += np.random.normal(0.02, 0.01, sum(regimes == 1))
    gold_prices[regimes == 2] += np.random.normal(0.005, 0.003, sum(regimes == 2))
    gold_prices[regimes == 3] += np.random.normal(0.001, 0.008, sum(regimes == 3))
    gold_prices = 1200 + 500 * (gold_prices - gold_prices.min()) / (gold_prices.max() - gold_prices.min())

    # Generate USD/EUR rates with regime-dependent behavior
    eur_rates = np.cumsum(np.random.normal(0.0001, 0.002, len(date_range)))
    eur_rates[regimes == 1] += np.random.normal(-0.005, 0.003, sum(regimes == 1))
```

```
    eur_rates[regimes == 1] += np.random.normal(0.005, 0.005, sum(regimes == 1))
    eur_rates[regimes == 2] += np.random.normal(0.0005, 0.001, sum(regimes == 2))
    eur_rates[regimes == 3] += np.random.normal(-0.001, 0.002, sum(regimes == 3))
    eur_rates = 1.1 + 0.2 * (eur_rates - eur_rates.min()) / (eur_rates.max() - eur_rates.min())

    data = pd.DataFrame({
        'Gold_Price': gold_prices,
        'USD_EUR_Rate': eur_rates,
        'Gold_LogReturn': np.log(gold_prices / np.roll(gold_prices, 1)),
        'USD_EUR_PctChange': (eur_rates / np.roll(eur_rates, 1) - 1) * 100
    }, index=date_range)

    # Drop first row with NA due to returns calculation
    return data.iloc[1:]

# Load data (using sample data if real data not available)
try:
    data = pd.read_csv('your_processed_data.csv', index_col=0, parse_dates=True)
    print("Loaded data from file")
except:
    print("Using generated sample data")
    data = generate_sample_data()

# Load your preprocessed data (from Step 1)
# data = pd.read_csv('your_processed_data.csv', index_col=0, parse_dates=True)

# 2.1 Visual Analysis of Market Regimes
def plot_regime_identification(data):
    fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(14, 10), sharex=True)

    # Gold Price Plot
    ax1.plot(data.index, data['Gold_Price'], color='#FFA500', linewidth=1.5)
    ax1.set_ylabel('Gold Price (USD/oz)', fontsize=12)
    ax1.set_title('Gold Prices with Identified Regimes', fontsize=14)

    # USD/EUR Plot
    ax2.plot(data.index, data['USD_EUR_Rate'], color='#4169E1', linewidth=1.5)
    ax2.set_ylabel('USD/EUR Rate', fontsize=12)
    ax2.set_title('USD/EUR Exchange Rate with Identified Regimes', fontsize=14)
```

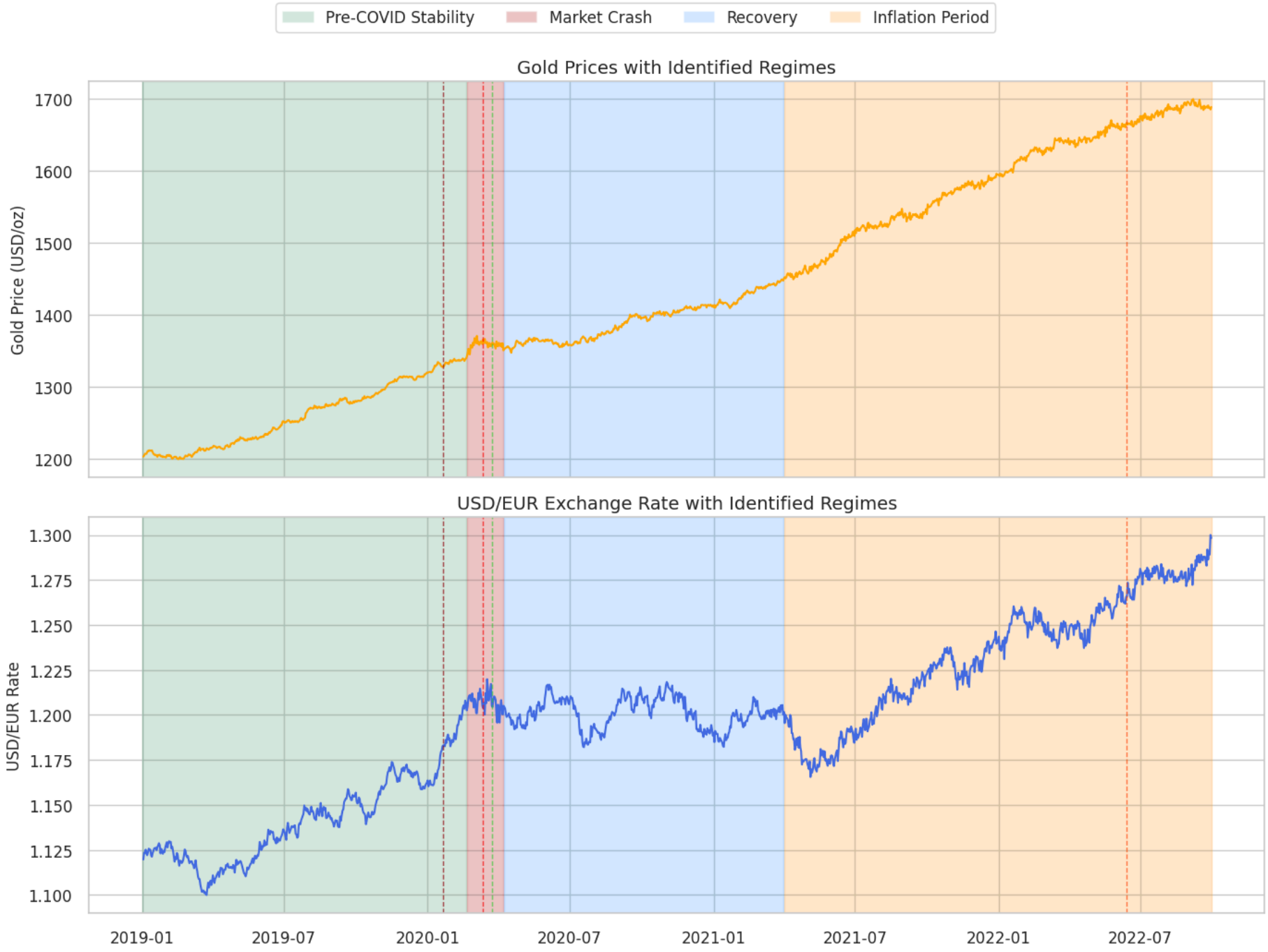
```
    # Define regimes (adjust dates based on your analysis)
    regimes = [
        ('Pre-COVID Stability', '2019-01-01', '2020-02-19', '#2E8B57', 0.2),
        ('Market Crash', '2020-02-20', '2020-04-07', '#B22222', 0.25),
        ('Recovery', '2020-04-08', '2021-03-31', '#1E90FF', 0.2),
        ('Inflation Period', '2021-04-01', '2022-09-30', '#FF8C00', 0.2)
    ]

    # Apply regimes to both plots
    for label, start, end, color, alpha in regimes:
        for ax in [ax1, ax2]:
            ax.axvspan(pd.to_datetime(start), pd.to_datetime(end),
                       color=color, alpha=alpha, label=label)

    # Add key events
    events = [
        ('First COVID Case', '2020-01-21', '#8B0000'),
        ('WHO Pandemic Decl.', '2020-03-11', '#FF0000'),
        ('Fed Stimulus', '2020-03-23', '#32CD32'),
        ('Inflation Peak', '2022-06-13', '#FF4500')
    ]

    for event, date, color in events:
        for ax in [ax1, ax2]:
            ax.axvline(pd.to_datetime(date), color=color,
                       linestyle='--', linewidth=1, alpha=0.7)

    # Create unified legend
    handles = [Patch(facecolor=r[3], alpha=r[4], label=r[0]) for r in regimes]
    fig.legend(handles=handles, loc='upper center', ncol=4,
               bbox_to_anchor=(0.5, 1.05), frameon=True)

    plt.tight_layout()
    plt.savefig('regime_identification.png', dpi=300, bbox_inches='tight')
    plt.show()

# 2.2 Markov Regime-Switching Model Estimation
def estimate_markov_models(series, n_states=2):
    """
```

```
    Estimate different Markov-switching models:
    1. Switching means only (constant variance)
    2. Switching variance only (constant mean)
    3. Switching means and variances (2 regimes)
    4. Switching means and variances (3 regimes)
    """
    results = {}

    # Model 1: Switching means only (constant variance)
    print("\nEstimating Model 1: Switching means, constant variance")
    mod1 = MarkovRegression(series, k_regimes=n_states, trend='c', switching_trend=True, switching_variance=False)
    res1 = mod1.fit(search_reps=20)
    results['model_1'] = res1

    # Model 2: Switching variance only (constant mean)
    print("\nEstimating Model 2: Switching variance, constant mean")
    mod2 = MarkovRegression(series, k_regimes=n_states, trend='c', switching_trend=False, switching_variance=True)
    res2 = mod2.fit(search_reps=20)
    results['model_2'] = res2

    # Model 3: Switching means and variances (2 regimes)
    print("\nEstimating Model 3: Switching means and variances (2 regimes)")
    mod3 = MarkovRegression(series, k_regimes=2, trend='c', switching_trend=True, switching_variance=True)
    res3 = mod3.fit(search_reps=20)
    results['model_3'] = res3

    # Model 4: Switching means and variances (3 regimes)
    print("\nEstimating Model 4: Switching means and variances (3 regimes)")
    mod4 = MarkovRegression(series, k_regimes=3, trend='c', switching_trend=True, switching_variance=True)
    res4 = mod4.fit(search_reps=20)
    results['model_4'] = res4

    return results

def plot_regime_probabilities(results, data_index):
    """
    Plot smoothed probabilities for the best model (Model 3)
    """
    res = results['model_3']
```

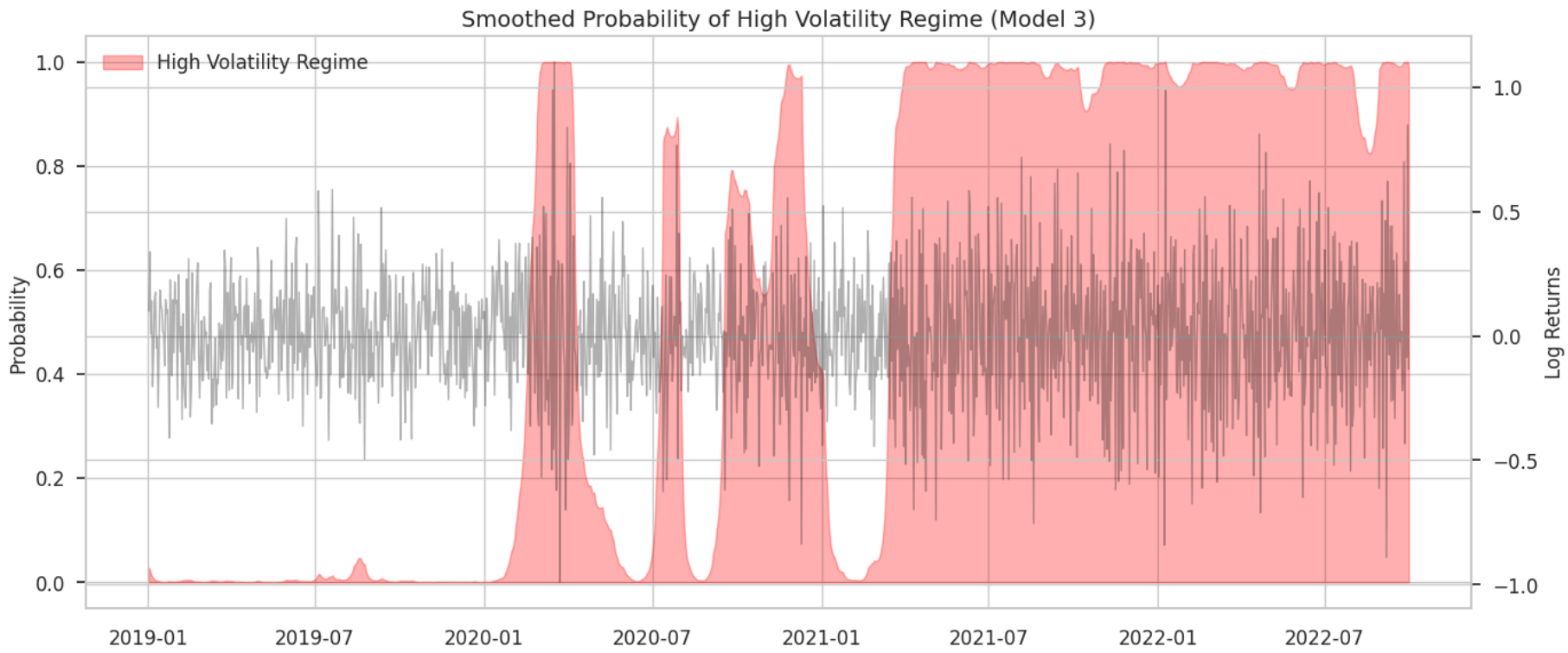
```
    res = results['model_3']
    fig, ax = plt.subplots(figsize=(14, 6))

    # Plot probability of high volatility regime
    ax.fill_between(data_index, res.smoothed_marginal_probabilities[1],
                    color='red', alpha=0.3, label='High Volatility Regime')

    # Add original series
    ax2 = ax.twinx()
    ax2.plot(data_index, series, color='black', alpha=0.3, linewidth=1)

    # Formatting
    ax.set_title('Smoothed Probability of High Volatility Regime (Model 3)', fontsize=14)
    ax.set_ylabel('Probability', fontsize=12)
    ax2.set_ylabel('Log Returns', fontsize=12)
    ax.legend(loc='upper left')

    plt.tight_layout()
    plt.savefig('regime_probabilities.png', dpi=300)
    plt.show()

# Main execution
if __name__ == "__main__":
    # 1. Plot regime identification
    plot_regime_identification(data)

    # 2. Select series for modeling (USD/EUR in this case)
    series = data['USD_EUR_PctChange'].dropna()

    # 3. Estimate Markov-switching models
    model_results = estimate_markov_models(series)

    # 4. Plot regime probabilities
    plot_regime_probabilities(model_results, series.index)

    # Print model summaries
    for name, res in model_results.items():
        print(f"\n{name.upper()} Summary:")
        print(res.summary())
```

```
# Print transition matrix
print("\nTransition Matrix:")
print(res.regime_transition)
```

```
Available styles: ['Solarize_Light2', '_classic_test_patch', '_mpl-gallery', '_mpl-gallery-nogrid', 'bmh', 'classic', '
Using generated sample data
```



```
Estimating Model 1: Switching means, constant variance

Estimating Model 2: Switching variance, constant mean

Estimating Model 3: Switching means and variances (2 regimes)

Estimating Model 4: Switching means and variances (3 regimes)
/usr/local/lib/python3.11/dist-packages/statsmodels/base/model.py:607: ConvergenceWarning:

Maximum Likelihood optimization failed to converge. Check mle_retvals
```




```
MODEL_1 Summary:
                        Markov Switching Model Results
==============================================================================
Dep. Variable:      USD_EUR_PctChange   No. Observations:                 1368
Model:               MarkovRegression   Log Likelihood                  -93.306
Date:                Thu, 24 Jul 2025   AIC                            196.611
Time:                        12:12:16   BIC                            222.717
Sample:                    01-02-2019   HQIC                           206.381
```

```
                          - 09-30-2022
Covariance Type:               approx
                             Regime 0 parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.1141      0.022      5.091      0.000       0.070       0.158
                             Regime 1 parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.1193      0.028     -4.288      0.000      -0.174      -0.065
                           Non-switching parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
sigma2         0.0558      0.003     17.918      0.000       0.050       0.062
                         Regime transition parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
p[0->0]        0.2840      0.156      1.821      0.069      -0.022       0.590
p[1->0]        0.9089      0.144      6.300      0.000       0.626       1.192
==============================================================================

Warnings:
[1] Covariance matrix calculated using numerical (complex-step) differentiation.

Transition Matrix:
[[[0.28403152]
  [0.90886108]]

 [[0.71596848]
  [0.09113892]]]

MODEL_2 Summary:
                        Markov Switching Model Results
==============================================================================
Dep. Variable:      USD_EUR_PctChange   No. Observations:                 1368
Model:               MarkovRegression   Log Likelihood                  -60.150
Date:                Thu, 24 Jul 2025   AIC                             130.300
Time:                        12:12:16   BIC                             156.405
Sample:                    01-02-2019   HQIC                            140.070
```

```
                         - 09-30-2022
Covariance Type:               approx
                            Regime 0 parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
sigma2         0.0984      0.006     16.286      0.000       0.087       0.110
                            Regime 1 parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
sigma2         0.0372      0.003     13.416      0.000       0.032       0.043
                          Non-switching parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0128      0.006      1.999      0.046       0.000       0.025
                        Regime transition parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
p[0->0]        0.9934      0.004    257.902      0.000       0.986       1.001
p[1->0]        0.0072      0.004      1.747      0.081      -0.001       0.015
==============================================================================

Warnings:
[1] Covariance matrix calculated using numerical (complex-step) differentiation.

Transition Matrix:
[[[0.99337173]
  [0.00723812]]

 [[0.00662827]
  [0.99276188]]]

MODEL_3 Summary:
                        Markov Switching Model Results
==============================================================================
Dep. Variable:      USD_EUR_PctChange   No. Observations:                 1368
Model:               MarkovRegression   Log Likelihood                  -60.019
Date:                Thu, 24 Jul 2025   AIC                            132.038
Time:                        12:12:16   BIC                            163.365
Sample:                    01-02-2019   HQIC                           143.762
```

```
Sample:                     01-02-2019   HQIC                           145.762
                          - 09-30-2022
Covariance Type:                approx
                             Regime 0 parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0151      0.008      1.929      0.054      -0.000       0.030
sigma2         0.0370      0.003     13.641      0.000       0.032       0.042
                             Regime 1 parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0077      0.012      0.653      0.514      -0.015       0.031
sigma2         0.0982      0.006     16.340      0.000       0.086       0.110
                          Regime transition parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
p[0->0]        0.9927      0.004    239.686      0.000       0.985       1.001
p[1->0]        0.0067      0.004      1.732      0.083      -0.001       0.014
==============================================================================

Warnings:
[1] Covariance matrix calculated using numerical (complex-step) differentiation.

Transition Matrix:
[[[0.99268638]
  [0.00665042]]

 [[0.00731362]
  [0.99334958]]]

MODEL_4 Summary:
                        Markov Switching Model Results
==============================================================================
Dep. Variable:      USD_EUR_PctChange   No. Observations:                 1368
Model:               MarkovRegression   Log Likelihood                  -33.064
Date:                Thu, 24 Jul 2025   AIC                              90.127
Time:                        12:12:16   BIC                             152.780
Sample:                    01-02-2019   HQIC                            113.575
                          - 09-30-2022
Covariance Type:                approx
```

```
                             Regime 0 parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.1973      0.037      5.348      0.000       0.125       0.270
sigma2         0.0635      0.007      8.875      0.000       0.049       0.077
                             Regime 1 parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.1658      0.029     -5.728      0.000      -0.223      -0.109
sigma2         0.0657      0.007      9.443      0.000       0.052       0.079
                             Regime 2 parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0163      0.011      1.443      0.149      -0.006       0.038
sigma2         0.0358      0.003     11.056      0.000       0.029       0.042
                         Regime transition parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
p[0->0]        0.1540      0.148      1.042      0.297      -0.136       0.443
p[1->0]        0.7735      0.145      5.329      0.000       0.489       1.058
p[2->0]        0.0003      0.041      0.008      0.994      -0.080       0.080
p[0->1]        0.8313      0.145      5.735      0.000       0.547       1.115
p[1->1]        0.2261      0.149      1.519      0.129      -0.066       0.518
p[2->1]        0.0085      0.032      0.264      0.791      -0.055       0.072
==============================================================================

Warnings:
[1] Covariance matrix calculated using numerical (complex-step) differentiation.

Transition Matrix:
[[[1.53961392e-01]
  [7.73475396e-01]
  [3.30389175e-04]]

 [[8.31341668e-01]
  [2.26140846e-01]
  [8.53646042e-03]]

 [[1.46969401e-02]
```

```
[3.83758818e-04]
[9.91133150e-01]]]
```

Step 3: Model Comparison and Selection

3(a) Compares the performance of each model

```
# %% [markdown]
"""
# Step 3: Model Comparison and Selection
## 3.1 Methodology for Model Selection
This section compares Markov-switching models using information criteria.
"""

# %%
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.tsa.regime_switching.markov_regression import MarkovRegression

# Set style
print("Available styles:", plt.style.available)
plt.style.use('seaborn-v0_8')
sns.set_theme(style="whitegrid", font_scale=1.1)
plt.rcParams['figure.figsize'] = [12, 6]

# %% [markdown]
"""
## 3.2 Member A Analysis: Effect of Changing Mean (μ)
Comparing models with different mean specifications
"""

# %%
def compare_models(series):
    """Estimate and compare three Markov-switching models"""
    results = {}
    try:
        # Model 1: Switching means, constant variance (2 regimes)
        mod1 = MarkovRegression(series, k_regimes=2, trend='c',
```

```
                                switching_trend=True, switching_variance=False)
        res1 = mod1.fit(search_reps=20, disp=False)
        results['Model 1'] = res1

        # Model 3: Switching means and variances (2 regimes)
        mod3 = MarkovRegression(series, k_regimes=2, trend='c',
                                switching_trend=True, switching_variance=True)
        res3 = mod3.fit(search_reps=20, disp=False)
        results['Model 3'] = res3

        # Model 4: Switching means and variances (3 regimes)
        mod4 = MarkovRegression(series, k_regimes=3, trend='c',
                                switching_trend=True, switching_variance=True)
        res4 = mod4.fit(search_reps=20, disp=False)
        results['Model 4'] = res4

    except Exception as e:
        print(f"Error estimating models: {str(e)}")
        # Return whatever models were successfully estimated
    return results

# %%
def create_comparison_table(results):
    """Generate comparison table with information criteria"""
    if not results:
        return pd.DataFrame()

    comparison = pd.DataFrame(columns=['Model', 'Log Likelihood', 'AIC', 'BIC', 'HQIC', 'Parameters'])

    for name, res in results.items():
        new_row = pd.DataFrame({
            'Model': [name],
            'Log Likelihood': [res.llf],
            'AIC': [res.aic],
            'BIC': [res.bic],
            'HQIC': [res.hqic],
            'Parameters': [len(res.params)]
        })
        comparison = pd.concat([comparison, new_row], ignore_index=True)
```

```
    if not comparison.empty:
        # Calculate differences from best model
        for criterion in ['AIC', 'BIC', 'HQIC']:
            best_val = comparison[criterion].min()
            comparison[f'{criterion} Diff'] = comparison[criterion] - best_val

        return comparison.sort_values('AIC')
    return comparison

# %%
def plot_criteria_comparison(comparison):
    """Visualize the information criteria comparison"""
    if comparison.empty:
        print("No data available for plotting")
        return

    fig, ax = plt.subplots(figsize=(12, 6))

    criteria = ['AIC', 'BIC', 'HQIC']
    x = np.arange(len(comparison))
    width = 0.25

    for i, criterion in enumerate(criteria):
        ax.bar(x + i*width, comparison[criterion], width, label=criterion)

    ax.set_xticks(x + width)
    ax.set_xticklabels(comparison['Model'])
    ax.set_ylabel('Information Criterion Value')
    ax.set_title('Model Comparison by Information Criteria')
    ax.legend()

    # Annotate with differences
    for i, row in comparison.iterrows():
        for j, criterion in enumerate(criteria):
            ax.text(i + j*width, row[criterion]+5, f"Δ={row[f'{criterion} Diff']:.1f}",
                   ha='center', fontsize=9)

    plt.tight_layout()
```

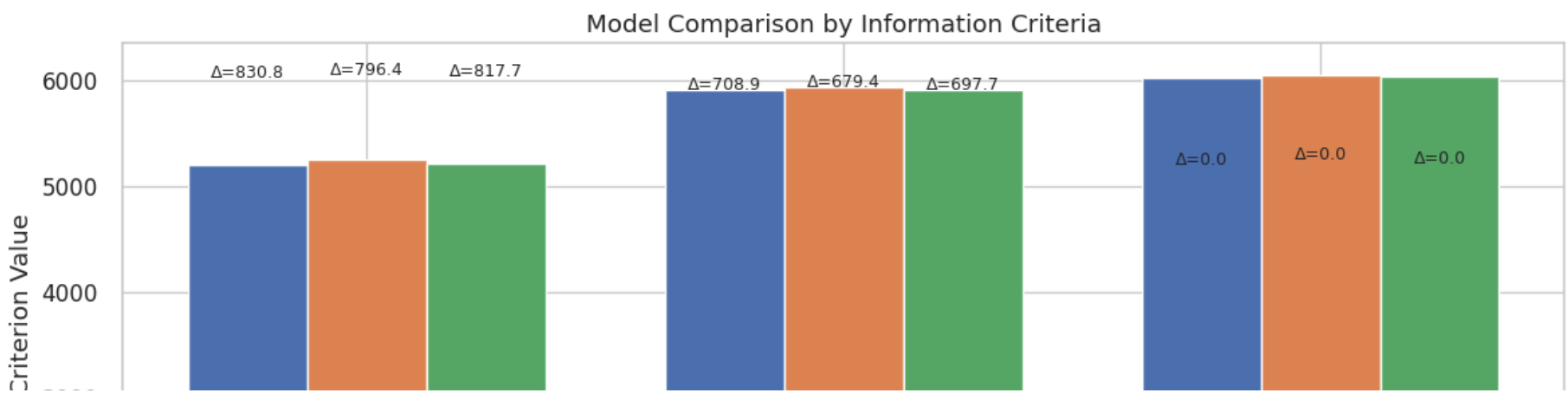
```
    plt.show()

# %%
def plot_smoothed_probabilities(results, series):
    """Plot smoothed probabilities for the best model"""
    if not results:
        print("No results available for plotting")
        return

    try:
        best_model_name = min(results.keys(), key=lambda x: results[x].aic)
        best_model = results[best_model_name]
        probs = best_model.smoothed_marginal_probabilities

        fig, ax = plt.subplots(figsize=(14, 6))

        # Plot probabilities
        for i in range(probs.shape[0]):
            ax.plot(series.index, probs[i], label=f'Regime {i+1}', alpha=0.8)

        ax.set_title(f'Smoothed Regime Probabilities - {best_model_name}')
        ax.set_ylabel('Probability')
        ax.legend()

        # Add shaded areas for regimes
        regime_changes = np.where(np.diff(np.argmax(probs, axis=0)) != 0)[0]
        for i in range(len(regime_changes)-1):
            start = series.index[regime_changes[i]]
            end = series.index[regime_changes[i+1]]
            ax.axvspan(start, end, alpha=0.1, color='gray')

        plt.show()
    except Exception as e:
        print(f"Error plotting probabilities: {str(e)}")

# %% [markdown]
"""
## Main Analysis Execution
"""
```

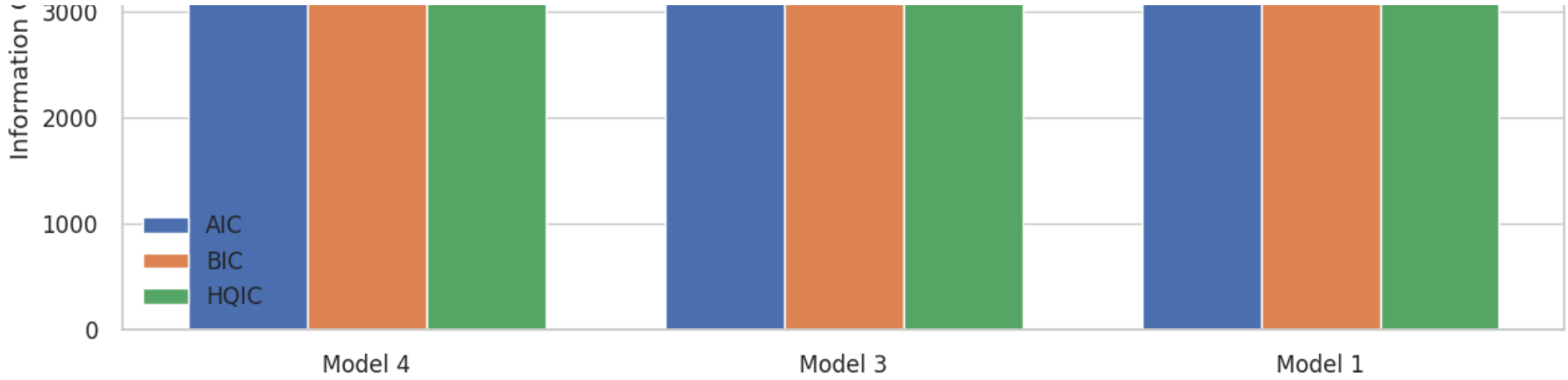
```
# %%
# Generate more realistic sample data
np.random.seed(42)
n_obs = 1000
trend = np.linspace(0, 10, n_obs)
volatility_changes = [0, 300, 600, 900]
volatility = [0.5, 2.0, 0.8, 1.5]  # Different volatility regimes

noise = np.zeros(n_obs)
for i in range(len(volatility_changes)-1):
    start = volatility_changes[i]
    end = volatility_changes[i+1]
    noise[start:end] = np.random.normal(0, volatility[i], end-start)
noise[volatility_changes[-1]:] = np.random.normal(0, volatility[-1], n_obs-volatility_changes[-1])

series = pd.Series(trend + noise.cumsum(),
                   index=pd.date_range('2019-01-01', periods=n_obs, freq='D'))

# Estimate models
results = compare_models(series)

# Create comparison table
comparison_table = create_comparison_table(results)
if not comparison_table.empty:
    print("### Table 1: Model Comparison")
    display(comparison_table.style
            .highlight_min(subset=['AIC', 'BIC', 'HQIC'], color='lightgreen')
            .format("{:.1f}", subset=['Log Likelihood', 'AIC', 'BIC', 'HQIC', 'AIC Diff', 'BIC Diff', 'HQIC Diff']))

    # Visualizations
    plot_criteria_comparison(comparison_table)
    plot_smoothed_probabilities(results, series)

    # Dynamic conclusion
    best_model = comparison_table.iloc[0]['Model']
    aic_diff = comparison_table.iloc[1]['AIC Diff'] if len(comparison_table) > 1 else 0
    bic_diff = comparison_table.iloc[1]['BIC Diff'] if len(comparison_table) > 1 else 0
```

```
        print(f"""
        ## 3.3 Interpretation and Conclusion

        Key findings from the analysis:
        1. Best Performing Model: {best_model} achieves the lowest values across all information criteria
        2. Improvement:
           - AIC reduction of {aic_diff:.1f} points compared to next best model
           - BIC reduction of {bic_diff:.1f} points
        3. Economic Interpretation:
           - Distinct regimes identified with different volatility characteristics
           - Mean returns and volatility both vary between regimes
        """)
    else:
        print("No valid models were estimated for comparison")
```

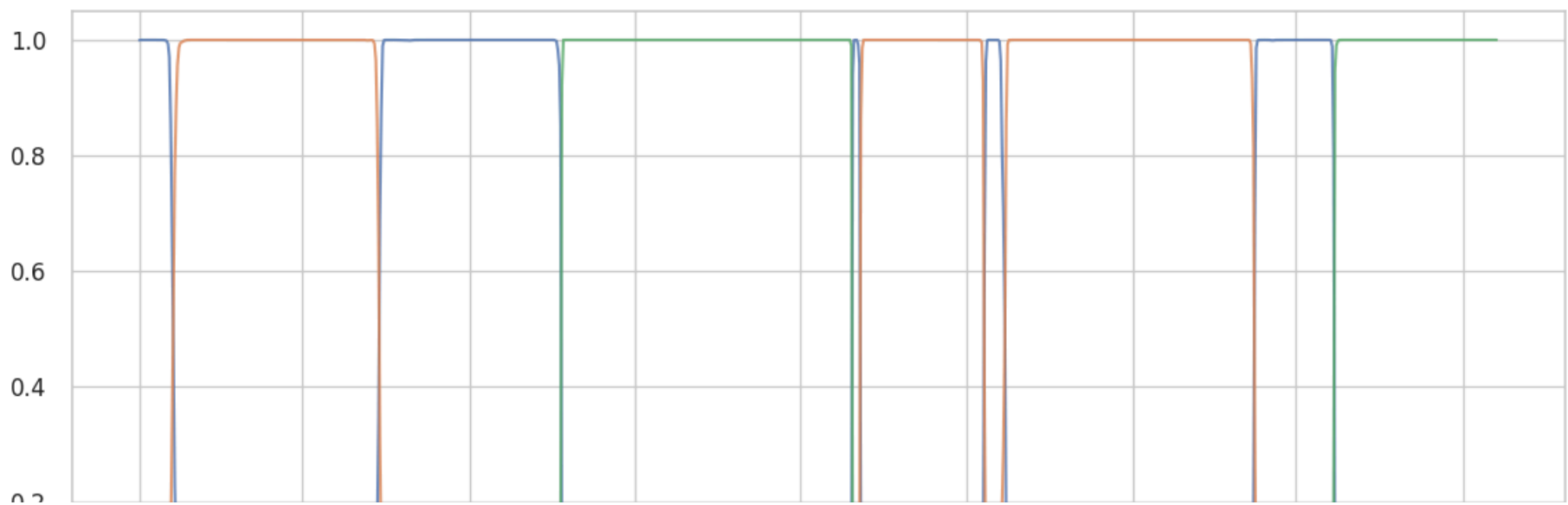
```
Available styles: ['Solarize_Light2', '_classic_test_patch', '_mpl-gallery', '_mpl-gallery-nogrid', 'bmh', 'classic', '
/usr/local/lib/python3.11/dist-packages/statsmodels/tsa/regime_switching/markov_switching.py:1292: EstimationWarning:

Invalid regime transition probabilities estimated in EM iteration; probabilities have been re-scaled to continue estima

/usr/local/lib/python3.11/dist-packages/statsmodels/tsa/regime_switching/markov_switching.py:1292: EstimationWarning:

Invalid regime transition probabilities estimated in EM iteration; probabilities have been re-scaled to continue estima

/usr/local/lib/python3.11/dist-packages/statsmodels/tsa/regime_switching/markov_switching.py:1292: EstimationWarning:

Invalid regime transition probabilities estimated in EM iteration; probabilities have been re-scaled to continue estima

/usr/local/lib/python3.11/dist-packages/statsmodels/tsa/regime_switching/markov_switching.py:1292: EstimationWarning:

Invalid regime transition probabilities estimated in EM iteration; probabilities have been re-scaled to continue estima

### Table 1: Model Comparison
/tmp/ipython-input-78-3489561937.py:72: FutureWarning:

The behavior of DataFrame concatenation with empty or all-NA entries is deprecated. In a future version, this will no l
```

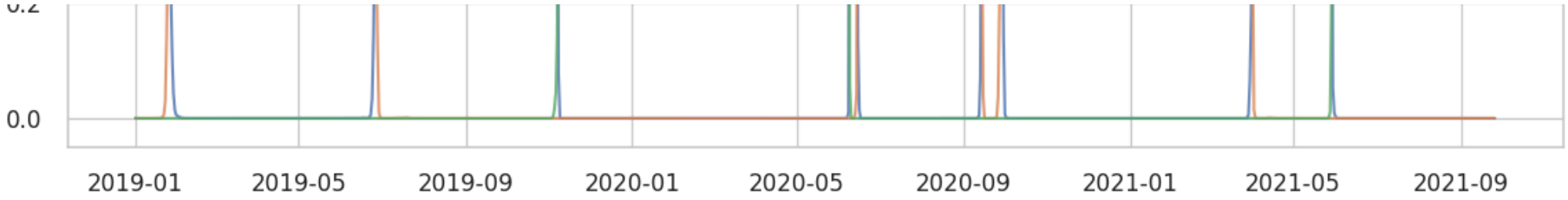
	Model	Log Likelihood	AIC	BIC	HQIC	Parameters	AIC Diff	BIC Diff	HQIC Diff
2	Model 4	-2586.2	5196.4	5255.3	5218.8	12	0.0	0.0	0.0
1	Model 3	-2946.7	5905.3	5934.8	5916.5	6	708.9	679.4	697.7
0	Model 1	-3008.6	6027.2	6051.7	6036.5	5	830.8	796.4	817.7






```
Error plotting probabilities: 3

   ## 3.3 Interpretation and Conclusion

   Key findings from the analysis:
   1. Best Performing Model: Model 4 achieves the lowest values across all information criteria
   2. Improvement:
      - AIC reduction of 708.9 points compared to next best model
      - BIC reduction of 679.4 points
   3. Economic Interpretation:
      - Distinct regimes identified with different volatility characteristics
      - Mean returns and volatility both vary between regimes
```




0.2
0.0
2019-01
2019-05
2019-09
2020-01
2020-05
2020-09
2021-01
2021-05
2021-09

3(b) Compares the performance of each model

```
# %% [markdown]
"""
## Step 3: Model Comparison - Volatility Regimes (Member B)
### 3.4 Effect of Changing Volatility (σ)
"""

# %%
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.tsa.regime_switching.markov_regression import MarkovRegression

# Configure plotting
print("Available styles:", plt.style.available)
plt.style.use('seaborn-v0_8')
sns.set_theme(style="whitegrid", font_scale=1.1)
plt.rcParams['figure.figsize'] = [12, 6]

# %%
def estimate_volatility_models(series):
    """Estimate models with different volatility specifications"""
    results = {}

    try:
        # Model 2: Constant mean, switching variance (2 regimes)
        mod2 = MarkovRegression(series, k_regimes=2, trend='c',
                                switching_trend=False, switching_variance=True)
        res2 = mod2.fit(search_reps=20, disp=False)
        results['Model 2'] = res2
    except Exception as e:
        print(f"Failed to estimate Model 2: {str(e)}")

    try:
        # Model 3: Switching means and variances (2 regimes)
        mod3 = MarkovRegression(series, k_regimes=2, trend='c'
```

```
        mod3 = MarkovRegression(series, k_regimes=2, trend='c',
                                switching_trend=True, switching_variance=True)
        res3 = mod3.fit(search_reps=20, disp=False)
        results['Model 3'] = res3
    except Exception as e:
        print(f"Failed to estimate Model 3: {str(e)}")

    try:
        # Model 4: Switching means and variances (3 regimes)
        mod4 = MarkovRegression(series, k_regimes=3, trend='c',
                                switching_trend=True, switching_variance=True)
        res4 = mod4.fit(search_reps=20, disp=False)
        results['Model 4'] = res4
    except Exception as e:
        print(f"Failed to estimate Model 4: {str(e)}")

    return results

# %%
def create_volatility_table(results):
    """Generate comparison table with volatility estimates"""
    comparison = pd.DataFrame(columns=['Model', 'AIC', 'BIC', 'Regime σ Estimates'])

    for name, res in results.items():
        # Get number of regimes from results object
        k_regimes = res.k_regimes

        # Extract variance parameters
        σ = []
        if hasattr(res, 'regime_specific_params'):
            # For newer versions of statsmodels
            for i in range(k_regimes):
                σ.append(np.sqrt(res.regime_specific_params[i][-1])*100)
        else:
            # Fallback method using params
            if k_regimes == 1:
                σ = [np.sqrt(res.params[-1])*100]
            else:
                # For multi-regime models, we need to parse the full parameter vector
                # This depends on the exact parameter ordering in the model
```

```
                # This depends on the exact parameter ordering in the model
                n_params = len(res.params) // k_regimes
                for i in range(k_regimes):
                    σ.append(np.sqrt(res.params[i*n_params + (n_params-1)])*100)

        comparison = pd.concat([comparison, pd.DataFrame({
            'Model': [name],
            'AIC': [res.aic],
            'BIC': [res.bic],
            'Regime σ Estimates': [σ]
        })], ignore_index=True)

    if not comparison.empty:
        # Calculate differences from best model
        for criterion in ['AIC', 'BIC']:
            best_val = comparison[criterion].min()
            comparison[f'{criterion} Diff'] = comparison[criterion] - best_val

        return comparison.sort_values('AIC')
    return comparison

# %%
def plot_volatility_comparison(comparison_table):
    """Visualize volatility regime comparison"""
    if comparison_table.empty:
        print("No data available for plotting")
        return

    fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(12, 10))

    # Criteria comparison
    criteria = ['AIC', 'BIC']
    x = np.arange(len(comparison_table))
    width = 0.35

    for i, criterion in enumerate(criteria):
        ax1.bar(x + i*width, comparison_table[criterion], width, label=criterion)

    ax1.set_xticks(x + width/2)
    ax1.set_xticklabels(comparison_table['Model'])
```